Ex. No. 1.1 – Design and Draft the Given 2D Sketches in Modelling Software

DATE:05.10.2026

AIM:

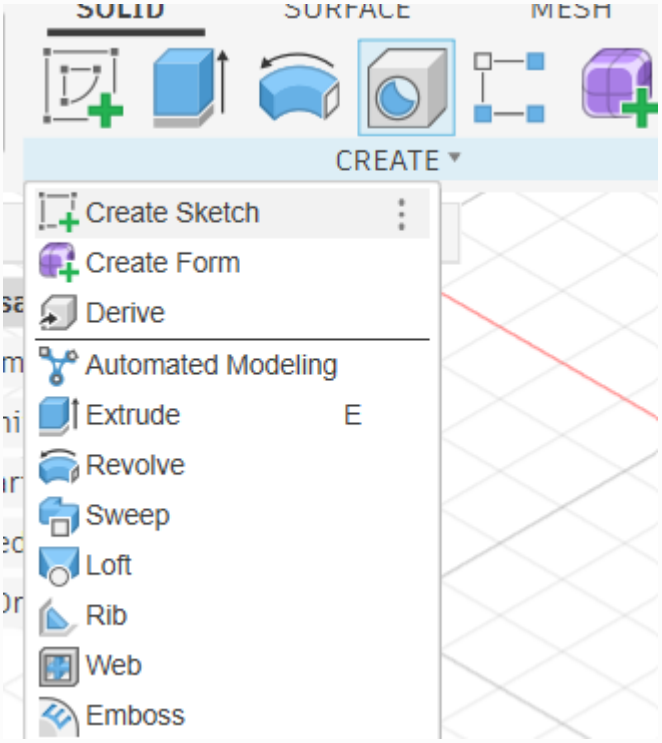
To sketch the given drawing with dimensions using Autodesk Fusion 360.

REQUIREMENTS:

- Autodesk Fusion 360
- Windows 10
- 16 GB of RAM (integrated graphics with 2 GB or more recommended)
- 2.5 Mbps or faster download; 500 Kbps or faster upload

PROCEDURE:

1. Open Autodesk Fusion 360 and create a new design file.
2. Select the **Create Sketch** option and choose a suitable plane for sketching.



3. Use sketching tools such as Line, Circle, Arc, Rectangle, and Polygon to draw the given 2D sketch.

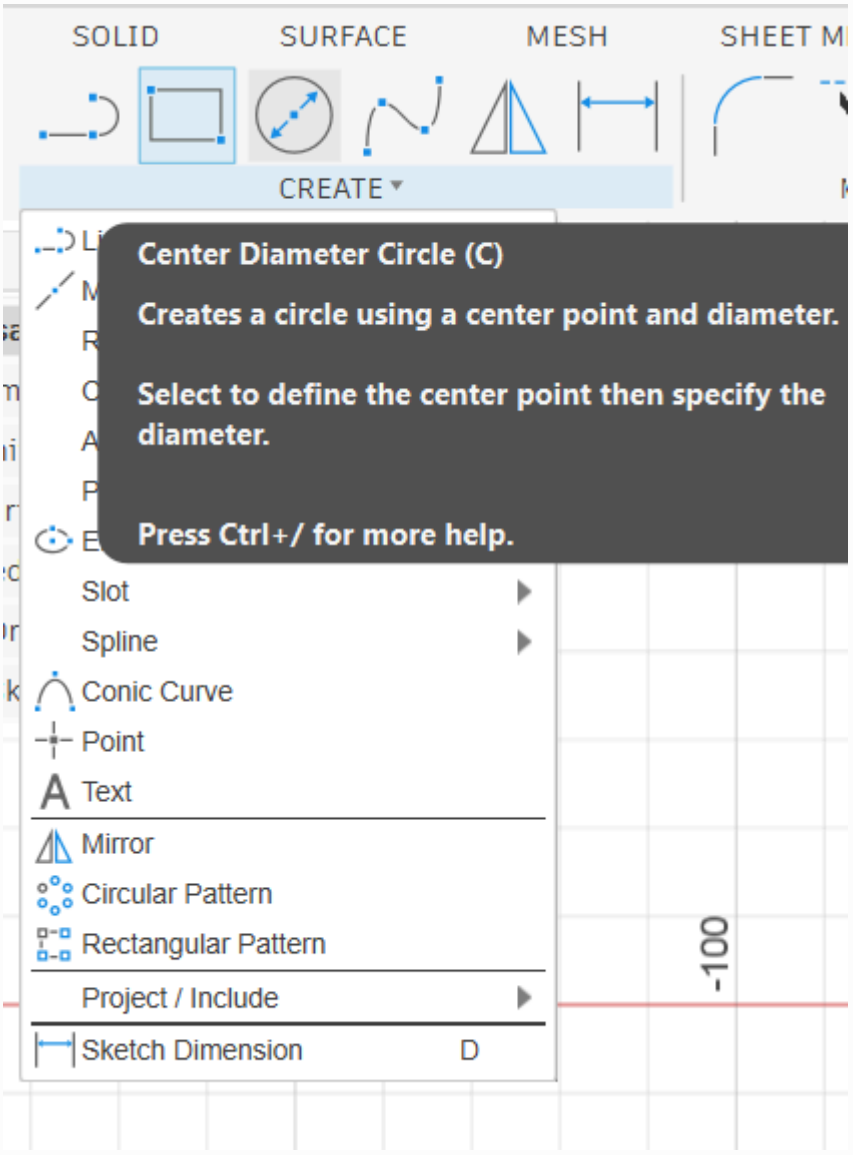


Image: Sketching tools and the initial sketch.

4. Apply the required dimensions using the **Sketch Dimension** command.

5. Use modifying tools such as Trim, Extend, Mirror, Move/Copy, and Fillet to edit the sketch as required.

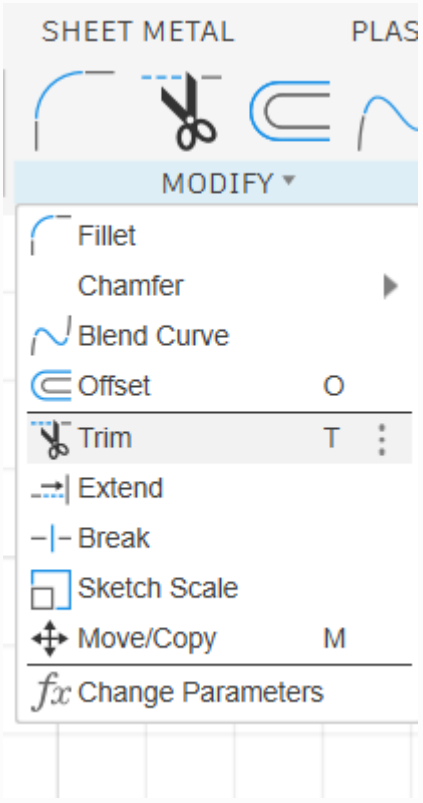


Image: Modified sketch using editing tools.

6. Apply suitable geometric constraints to maintain the correct shape and alignment.
7. Use the Inspect tools to check the dimensions and verify the sketch geometry.

Image: Inspection of sketch dimensions.

8. After completing the sketch, click **Finish Sketch** to finalize the 2D geometry.

Image: Completed 2D sketch.

9. Switch to the Drawing workspace and select **Drawing** → **From Design** to prepare the engineering drawing.

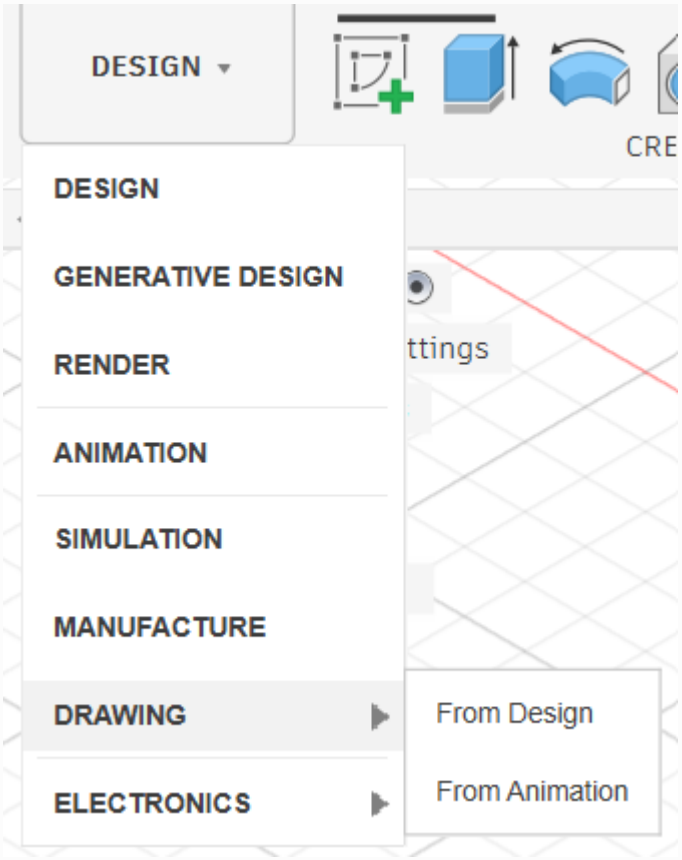


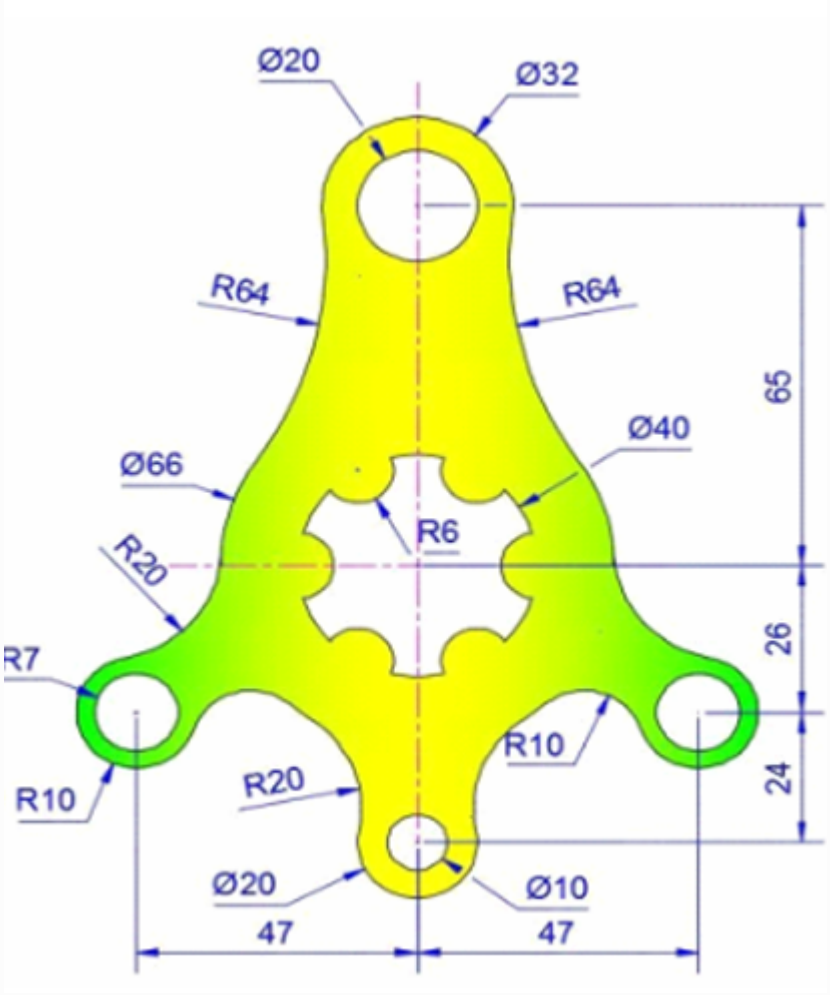
Image: Creating a drawing from the design.

10. Select the required sheet size, units, and drawing standard. Place the necessary views and add dimensions and annotations.

Image: Final dimensioned engineering drawing.

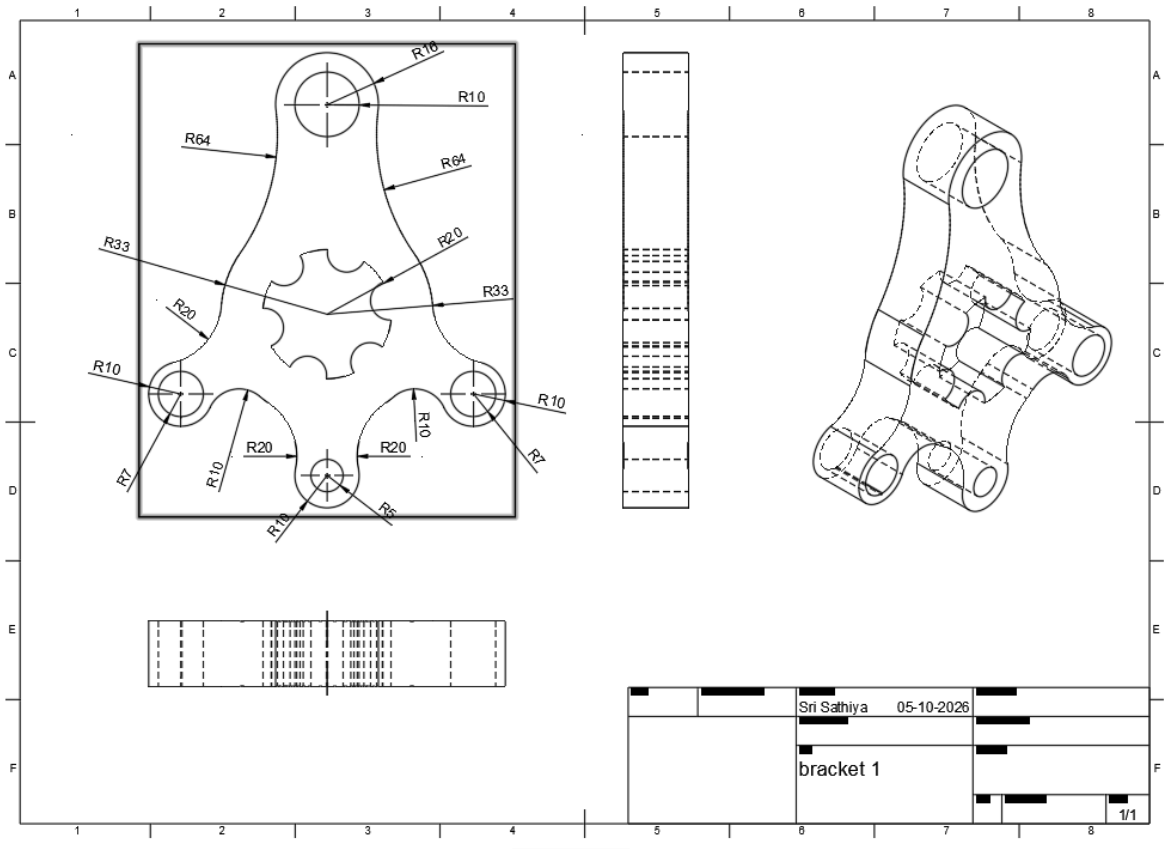
SKETCHES:

Insert the given 2D sketches and the corresponding completed sketches.



OUTPUT:

The given 2D sketches are drawn with the required dimensions and drafted using Autodesk Fusion 360.



RESULT:

Thus, the given 2D sketches were drawn and drafted successfully using Autodesk Fusion 360.